

Bayesian A/B Testing

Intro: CS 677 - Bayesian Methods in Computer Science

During my last semester of college at Brigham Young University, I enrolled in CS 677, a class on Bayesian methods in Computer Science. I had learned Bayes' rule in school previously, but I never quite understood what it meant to be a Bayesian. Porter Jenkins, a professor I had previously researched, was teaching the class. He and I had had many conversations about Bayesian methods during our time doing research together, so I was looking forward to this class. Unexpectedly, I came to absolutely love the theory and methodology taught in this class. It became one of my favorite classes at BYU.

The class started with fundamentals such as probability theory and built into complex topics such as conjugate priors, Monte Carlo sampling, variational inference, deep ensembles, and decision theory. While I am by no means an expert in all of these topics, I finished the class with a deep appreciation and understanding of conjugate priors and Bayesian decision theory. In this essay, I want to lay out an interesting approach to A/B testing that leverages both of those topics.

Background: Bayesian Theory

The core tenet of Bayesian theory is that probabilities represent a degree of belief or uncertainty about an event or model parameter, rather than a fixed long-run frequency. This idea is formalized in Bayes' rule:

$$p(A|B) = p(B|A)p(A) / p(B)$$

In words: the posterior equals the likelihood times the prior, divided by the marginal likelihood. The prior captures what we believe before seeing any data. The likelihood quantifies how probable the observed data are under a given hypothesis. The marginal likelihood is a normalization term that ensures the posterior is a valid probability distribution. Together, these produce the posterior — our prior belief updated in light of the data.

To make this concrete, suppose event A is "having COVID" and event B is "testing positive." Before taking the test, we believe a random person has a 1% chance of being infected, so our prior is $p(A)=0.01$. We also know two things about the test: if someone actually has COVID, it returns positive 90% of the time, so $p(B|A)=0.90$; and across the whole population, about 6% of tests come back positive, so $p(B)=0.06$. You take the test, and it's positive. What's the probability you actually have COVID?

We can calculate that using Bayes rule: $1\% \times 90\% / 6\% = 15\%$

So even after a positive result, there's only a 15% chance you actually have COVID. Your belief has shifted substantially — from 1% up to 15%, a 15x increase — but it's still well below 50% because the prior was so low to begin with. This is the classic base-rate result: a positive test from a fairly accurate diagnostic test isn't nearly as alarming as it intuitively feels, because most positives come from the much larger pool of healthy people.

Knowledge check: if you understand why this comic is funny, you're on track.

